

SIMATSENGINEERING

ASSESSMENTTOOL3

NAME: kamboji Thanuja
COURSE NAME: CSA16

REG.NO: 192373012
COURSE CODE: Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO4: Examine the applicability of clustering algorithms in real-world scenarios, presenting findings through clear reports with effective visualizations. Assessment Tool Weightage: 20%

QUESTIONS: 05

TotalMarks:50

Annexure A

COMPARITIVEANALYSIS

<i>Criteria</i>	Scope of Items (2)	Comparison Depth(2.5)	Use of Evidence (2)	Critical Thinking (2)	Clarity of Presentation (1)	<i>Total(10)</i>
<i>Weightage</i>	20%	25%	20%	20%	10%	<i>100%</i>
<i>Q1</i>						
<i>Q2</i>						
<i>Q3</i>						
<i>Q4</i>						
<i>Q5</i>						

Code of Conduct:

*I _k Thanuja/192373012 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

K Thanuja

RUBRICS FOR CALCULATION:

		Comparative analysis			
Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Scope of Items	20%	Selects highly relevant items with clear scope and justification	Selects appropriate items with minor gaps	Selection is limited or partially relevant	Items are unclear or not relevant
Comparison Depth	25%	Provides detailed and comprehensive comparison across multiple aspects	Provides adequate comparison with some depth	Limited comparison with few aspects	Very minimal or no comparison
Use of Evidence	20%	Supports comparison with accurate data, examples, or references	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Critical Thinking	20%	Demonstrates strong critical thinking with clear interpretation and reasoning	Shows reasonable interpretation with minor gaps	Basic interpretation; lacks depth	No meaningful interpretation
Clarity of Presentation	15%	Well-organized, clear, and logically structured presentation	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

Annexure A

Q1 . Dataset: Assessment Tool 3: Comparative Analysis (25% → 5 Marks)

Time duration :45 Minutes

Assessment Tool 3 (Low)

Q1 . Dataset: Customer Data **Customer**

Income (₹) Spending Score

C1	30000	60
C2	60000	40
C3	50000	70
C4	28000	65
C5	65000	35

Question:

Compare K-means and hierarchical clustering using this dataset and explain differences in performance and scalability. (BL3 – 1 Mark)

OUTPUT

The screenshot shows the Weka Explorer interface with the HierarchicalClusterer model selected. The 'Clusterer' tab is active, displaying the command: `HierarchicalClusterer -N 2 -L SINGLE -P -A "weka.core.EuclideanDistance -R first-last"`. The 'Cluster mode' section has 'Use training set' selected. The 'Clusterer output' pane shows the following details:

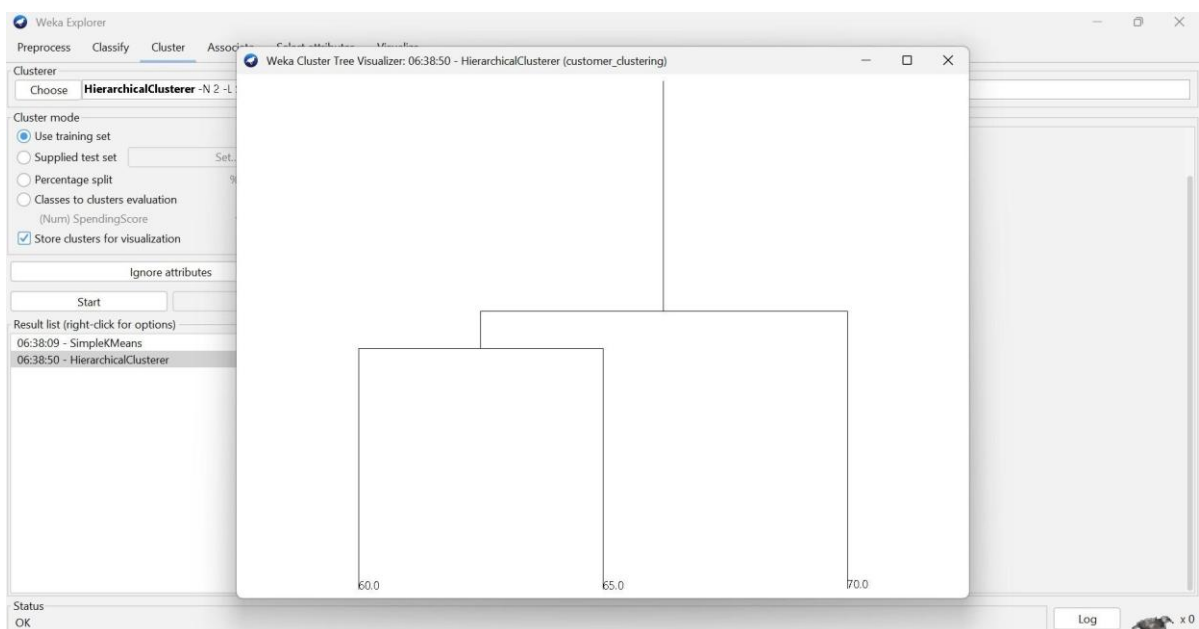
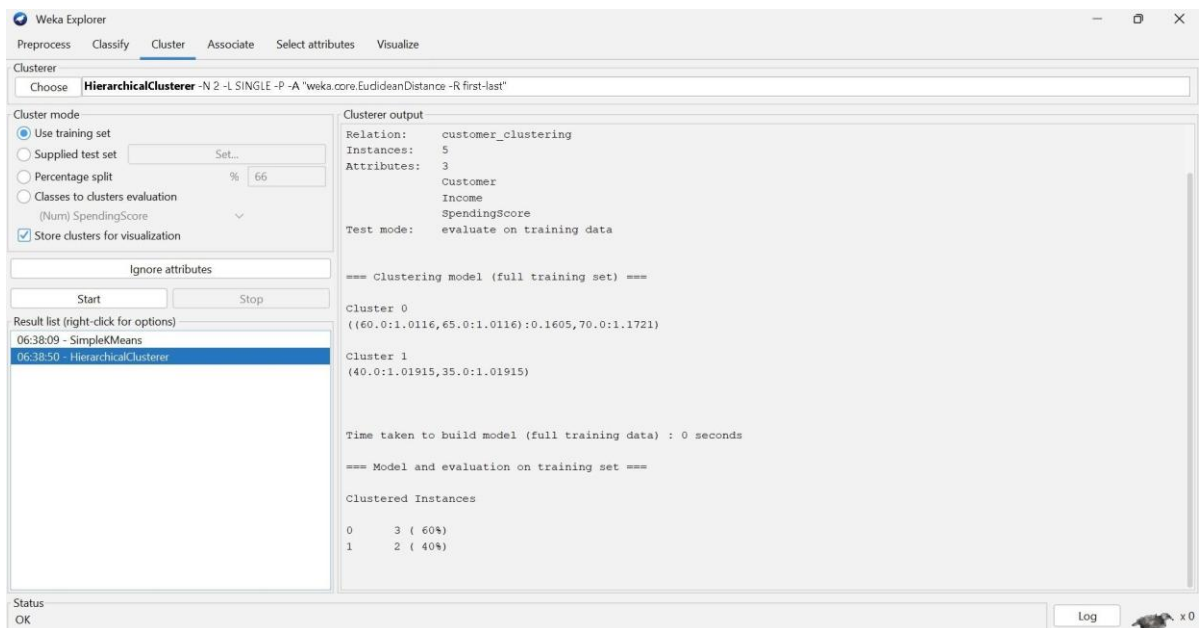
- Relation: customer_clustering
- Instances: 5
- Attributes: 3 (Customer, Income, SpendingScore)
- Test mode: evaluate on training data

The output text is as follows:

```
=== Clustering model (full training set) ===  
Cluster 0  
( (60.0:1.0116,65.0:1.0116):0.1605,70.0:1.1721)  
Cluster 1  
(40.0:1.01915,35.0:1.01915)  
  
Time taken to build model (full training data) : 0 seconds  
  
=== Model and evaluation on training set ===  
  
Clustered Instances  
0      3 ( 60%)  
1      2 ( 40%)
```

The 'Result list' on the left shows two entries: '06:38:09 - SimpleKMeans' and '06:38:50 - HierarchicalClusterer', with the latter selected.

This screenshot shows the same Weka Explorer interface as above, but with a 'Weka Cluster Tree Visualizer' window overlaid. The window title is 'Weka Cluster Tree Visualizer: 06:38:50 - HierarchicalClusterer (customer_clustering)'. It displays a hierarchical tree diagram representing the clustering process. The x-axis at the bottom is labeled with values 60.0, 65.0, and 70.0. The tree structure shows two main branches merging at the top, with further subdivisions below.



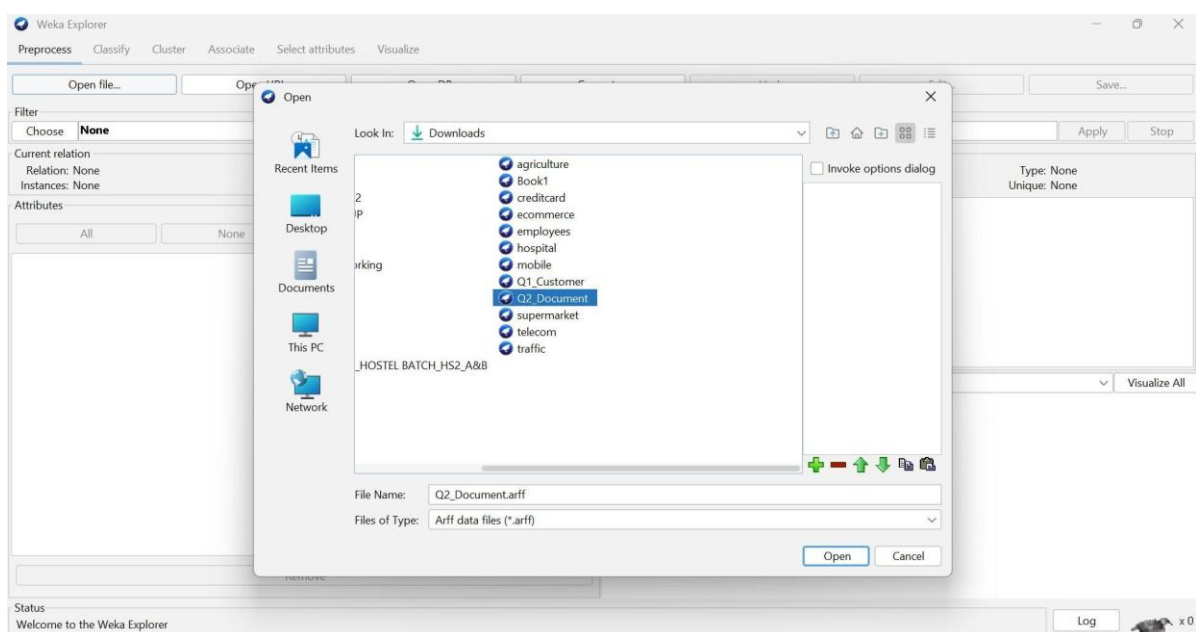
Q2. Dataset: Document Data

Document Word1 Word2 Word3

D1	0.2	0.5	0.3
D2	0.1	0.6	0.3
D3	0.8	0.1	0.1
D4	0.75	0.15	0.1
D5	0.2	0.4	0.4

Question:

Compare K-means and EM algorithms in handling overlapping clusters for the given dataset.
(BL4 – 1 Mark) OUTPUT:



Weka Explorer

Preprocess Classify **Cluster** Associate Select attributes Visualize

Clusterer Choose **SimpleKMeans** -init 0 -max-candidates 100 -periodic-pruning 10000 -min-density 2.0 -t1 -1.25 -t2 -1.0 -N 2 -A "weka.core.EuclideanDistance -R first-last" -I 500 -num-slots 1 -S 10

Cluster mode

☒ Use training set

☐ Supplied test set Set...

☐ Percentage split % 66

☐ Classes to clusters evaluation (Num) Word3

☒ Store clusters for visualization

Ignore attributes

Start Stop

Result list (right-click for options)

06:43:03 - SimpleKMeans

Clusterer output

Cluster 0: D4,0.75,0.15,0.1
Cluster 1: D2,0.1,0.6,0.3

Missing values globally replaced with mean/mode

Final cluster centroids:

Attribute	Full Data (5.0)	Cluster# 0 (2.0)	Cluster# 1 (3.0)
Document	D1	D3	D1
Word1	0.41	0.775	0.1667
Word2	0.35	0.125	0.5
Word3	0.24	0.1	0.3333

Time taken to build model (full training data) : 0.01 seconds

=== Model and evaluation on training set ===

Clustered Instances

0	2 (40%)
1	3 (60%)

Status OK Log x0

Weka Explorer

Preprocess Classify **Cluster** Associate Select attributes Visualize

Clusterer Choose **EM** -I 100 -N -1 -X 10 -max -1 -ll-cv 1.0E-6 -ll-iter 1.0E-6 -M 1.0E-6 -K 10 -num-slots 1 -S 100

Cluster mode

☒ Use training set

☐ Supplied test set Set...

☐ Percentage split % 66

☐ Classes to clusters evaluation (Num) Word3

☒ Store clusters for visualization

Ignore attributes

Start Stop

Result list (right-click for options)

06:43:03 - SimpleKMeans

06:43:27 - EM

Clusterer output

D4 4
D5 2
[total] 10
Word1
mean 0.41
std. dev. 0.3007
Word2
mean 0.35
std. dev. 0.1949
Word3
mean 0.24
std. dev. 0.12

Time taken to build model (full training data) : 0.01 seconds

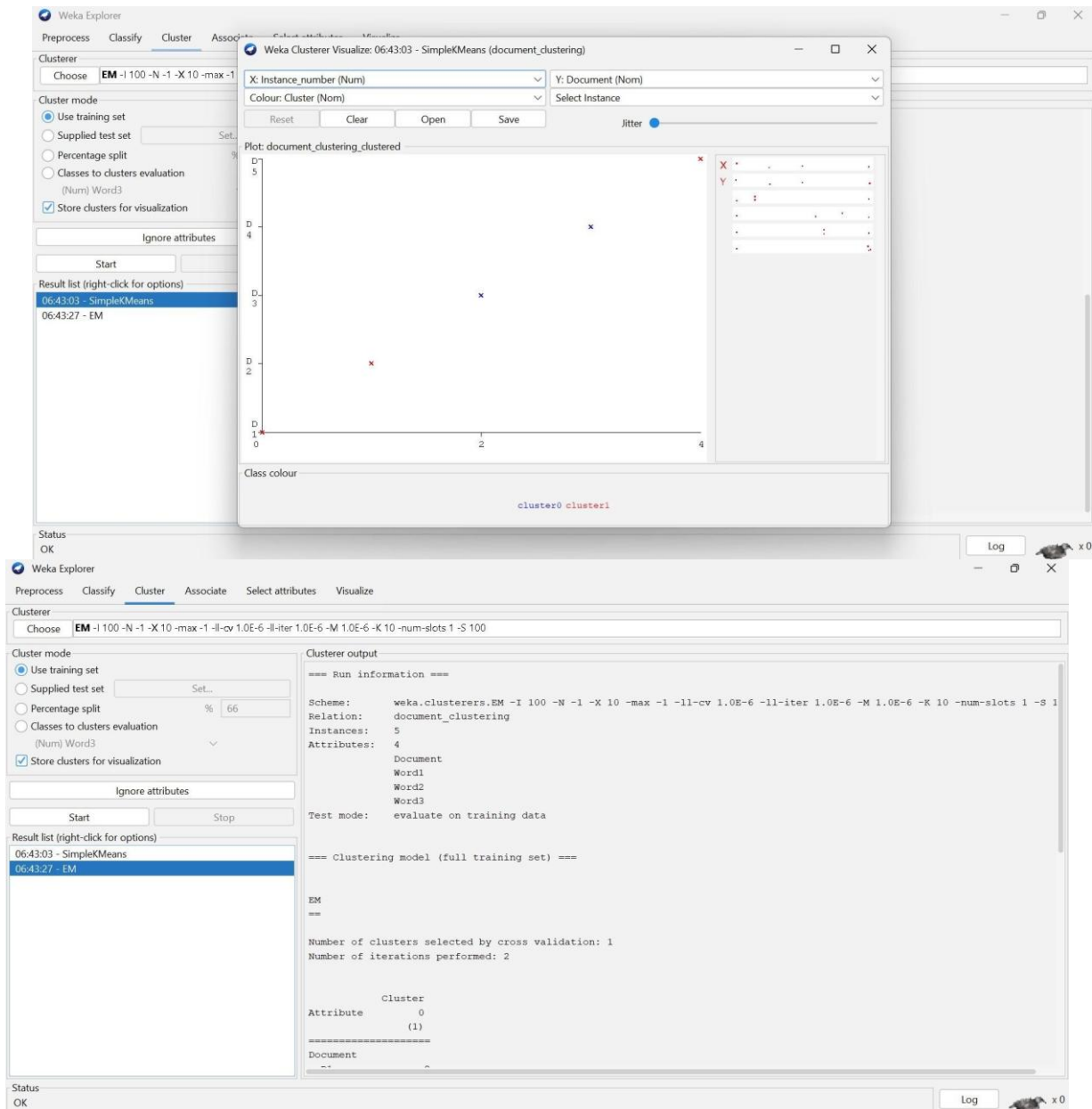
=== Model and evaluation on training set ===

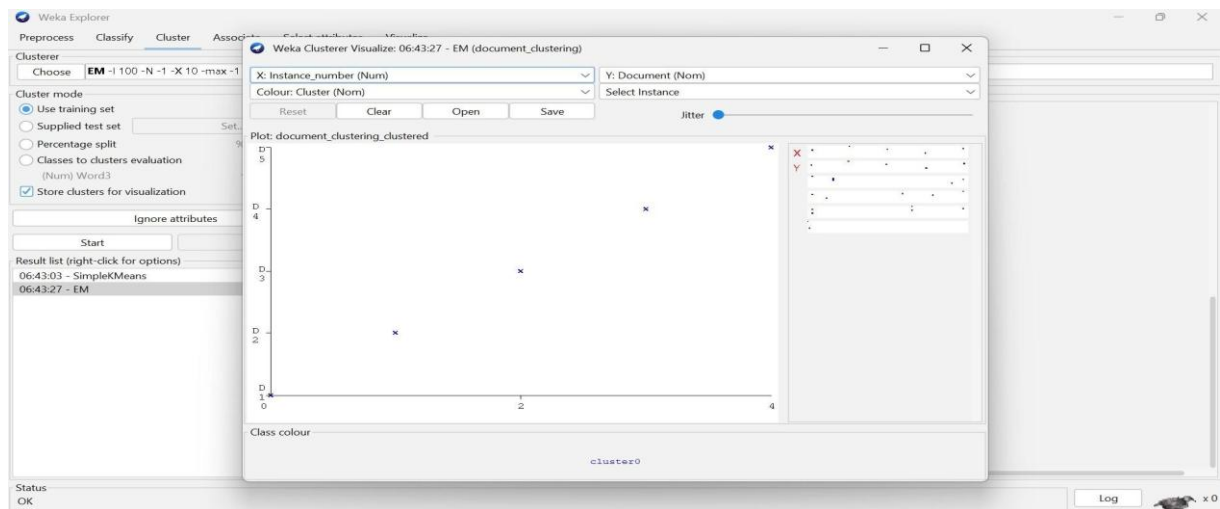
Clustered Instances

0	5 (100%)
---	----------

Log likelihood: -0.90915

Status OK Log x0





Q3. Dataset: Geographic Data

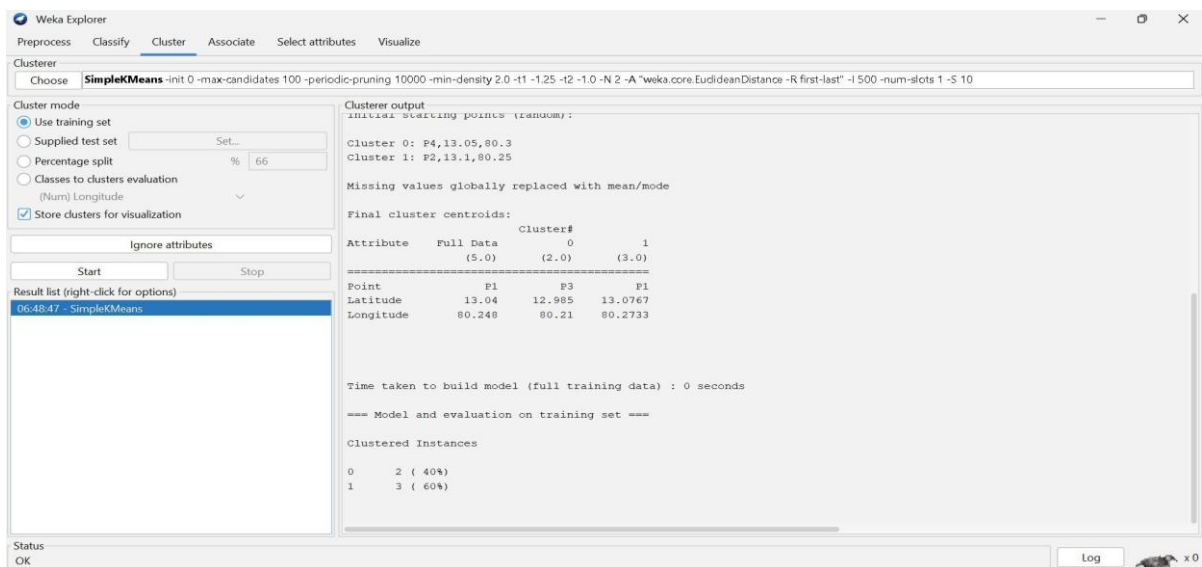
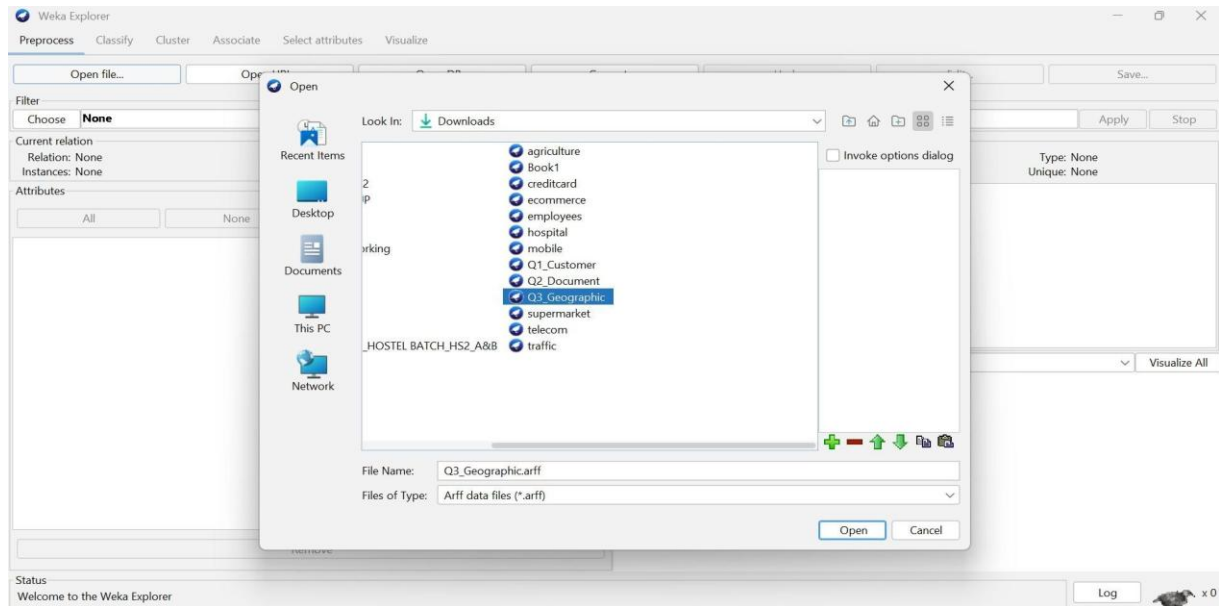
Point Latitude Longitude

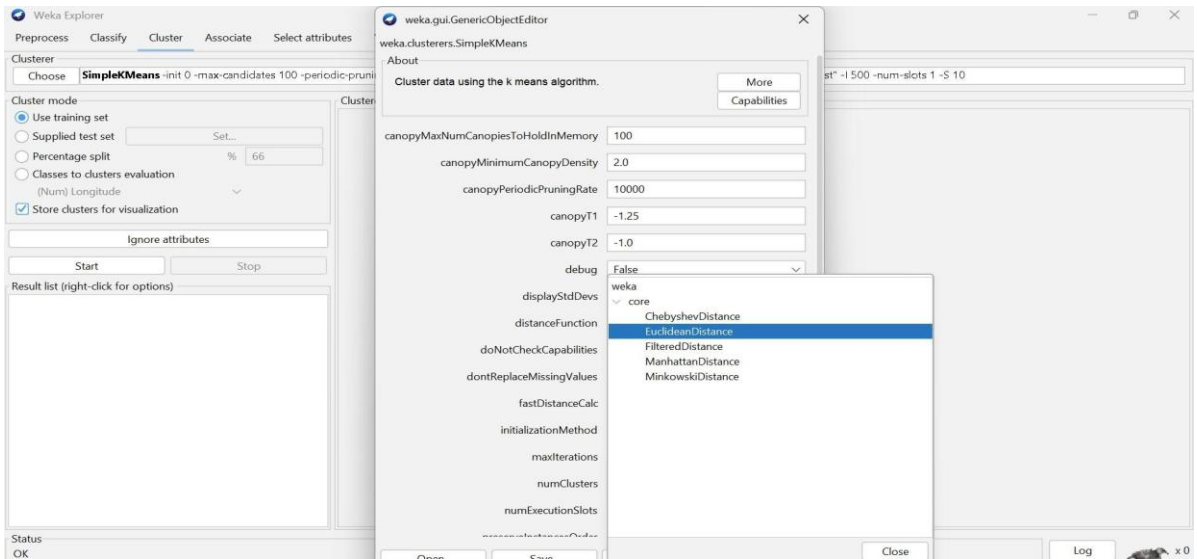
P1	13.08	80.27
P2	13.10	80.25
P3	12.97	80.22
P4	13.05	80.30
P5	13.00	80.20

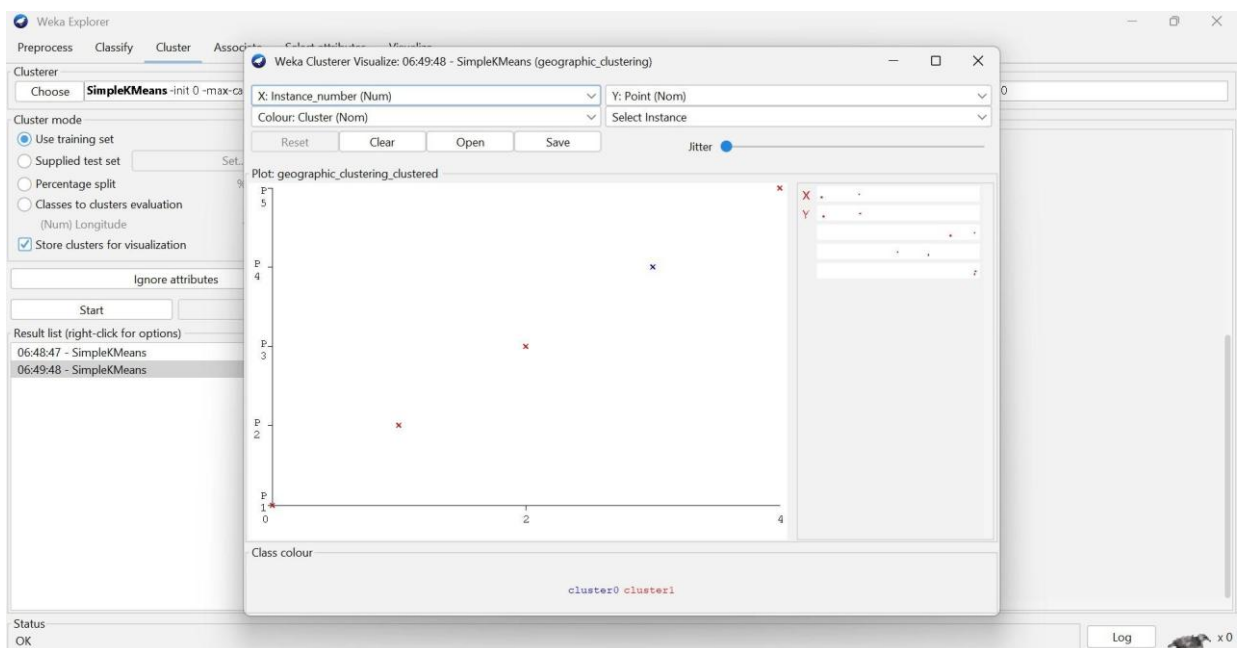
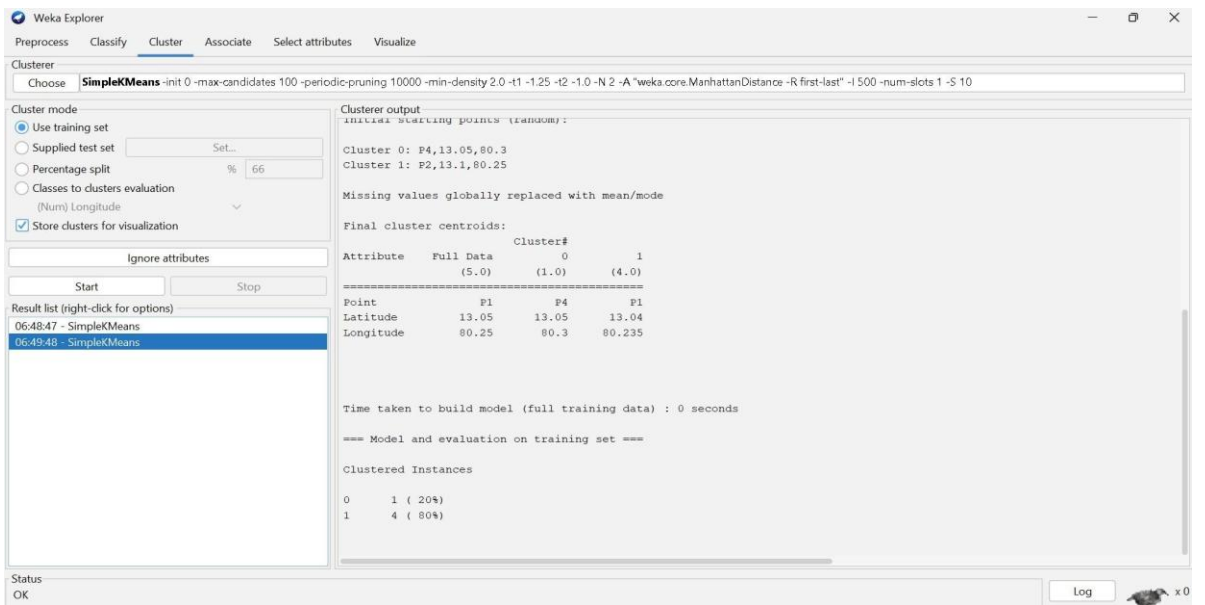
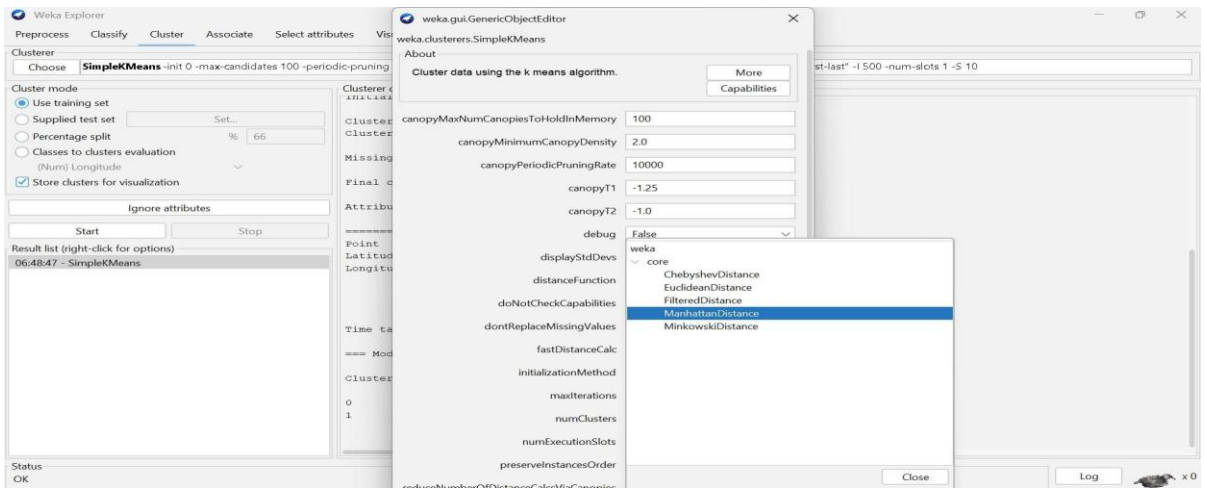
Question:

Compare Euclidean and Manhattan distance metrics and analyze their impact on clustering results. (BL4 – 1 Mark)

OUTPUT:







Q4. Dataset: Retail Products

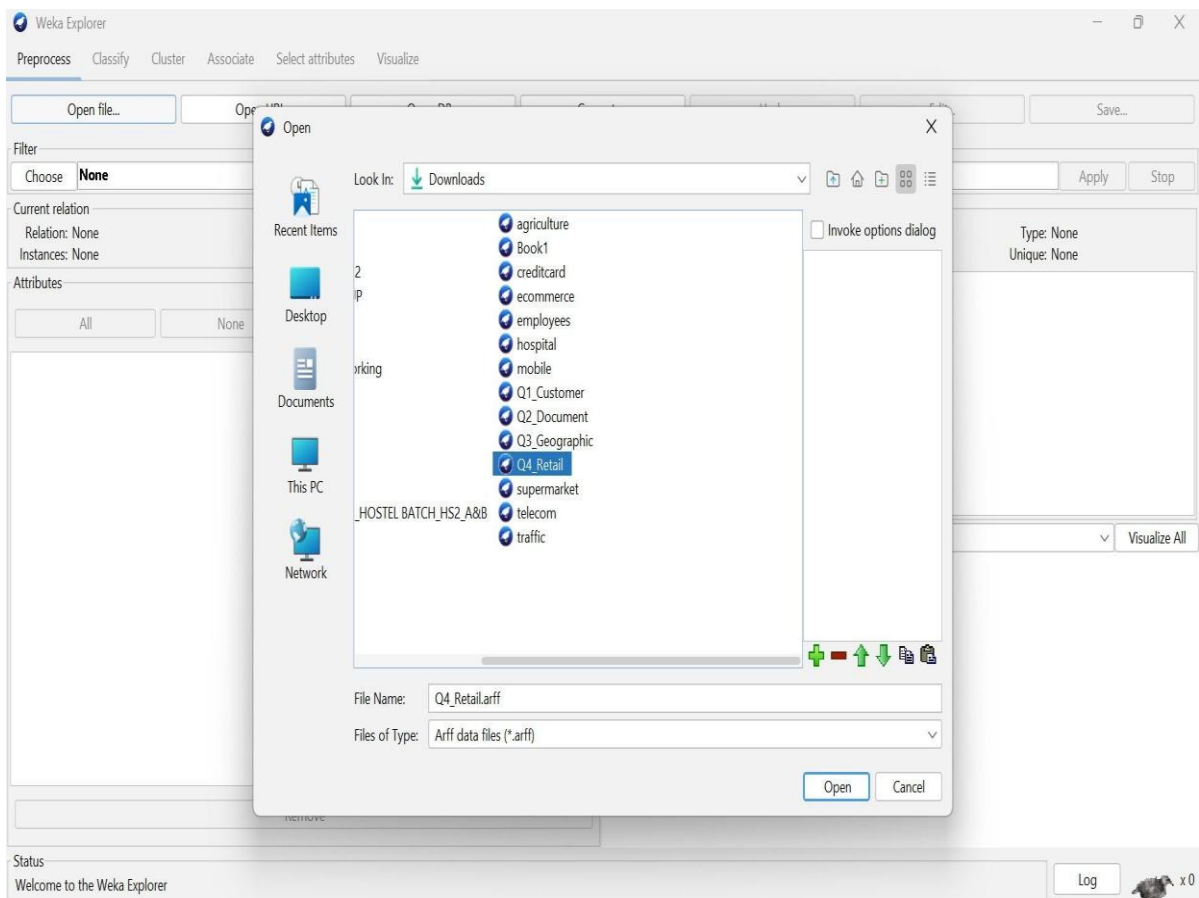
Product Sales (₹) Frequency

P1	50000	20
P2	30000	10
P3	52000	22
P4	25000	8
P5	32000	12

Question:

Compare agglomerative and divisive clustering approaches and explain their suitability for this dataset. (BL3 – 1 Mark)

OUTPUT:



Weka Explorer

Preprocess

Classify

Cluster

Associate

Select attributes

Visualize

Clusterer

ChooseHierarchicalClusterer -N 2 -L SINGLE -P -A "weka.core.EuclideanDistance -R first-last"

Cluster mode

☒ Use training set

☐ Supplied test set

☐ Percentage split

☐ Classes to clusters evaluation

☒ Store clusters for visualization

Set...

%66

(Num) Frequency

Ignore attributes

Start

Stop

Result list (right-click for options)

06:53:39 - HierarchicalClusterer

Cluster output

Relation: retail_products

Instances: 5

Attributes: 3

Product

Sales

Frequency

Test mode: evaluate on training data

=== Clustering model (full training set) ===

Cluster 0

(20.0:1.01286,22.0:1.01286)

Cluster 1

((10.0:1.01286,12.0:1.01286):0.01412,8.0:1.02699)

Time taken to build model (full training data) : 0 seconds

=== Model and evaluation on training set ===

Clustered Instances

02 (40%)

13 (60%)

Status

OK

Log

x0

Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Cluster

Choose **MakeDensityBasedClusterer** -M 1.0E-6 -W weka.clusterers.SimpleKMeans --init 0 -max-candidates 100 -periodic-pruning 10000 -min-density 2.0 -t1 -1.25 -t2 -1.0 -N 2 -A "weka.core.EuclideanDistance -R first-last" -I 500 -n

Cluster mode

☒ Use training set

☐ Supplied test set Set...

☐ Percentage split % 66

☐ Classes to clusters evaluation (Num) Frequency

☒ Store clusters for visualization

Ignore attributes

Start Stop

Result list (right-click for options)

06:53:39 - HierarchicalClusterer

06:54:50 - MakeDensityBasedClusterer

Cluster output

```
Discrete Estimator. Counts = 1 2 1 2 2 (Total = 8)
Attribute: Sales
Normal Distribution. Mean = 29000 StdDev = 2943.9203
Attribute: Frequency
Normal Distribution. Mean = 10 StdDev = 1.633

Cluster: 1 Prior probability: 0.4286

Attribute: Product
Discrete Estimator. Counts = 2 1 2 1 1 (Total = 7)
Attribute: Sales
Normal Distribution. Mean = 51000 StdDev = 1000
Attribute: Frequency
Normal Distribution. Mean = 21 StdDev = 1

Time taken to build model (full training data) : 0 seconds

=== Model and evaluation on training set ===

Clustered Instances

0    3 ( 60%)
1    2 ( 40%)

Log likelihood: -12.6953
```

Status OK

Log x 0

Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Cluster

Choose **MakeDensityBasedClusterer**

Cluster mode

☒ Use training set

☐ Supplied test set Set...

☐ Percentage split %

☐ Classes to clusters evaluation (Num) Frequency

☒ Store clusters for visualization

Ignore attributes

Start

Result list (right-click for options)

06:53:39 - HierarchicalClusterer

06:54:50 - MakeDensityBasedClusterer

Weka Clusterer Visualize: 06:54:50 - MakeDensityBasedClusterer (retail_products)

X: Instance_number (Num) Y: Product (Nom)

Colour: Cluster (Nom) Select Instance

Reset Clear Open Save Jitter

Plot: retail_products_clustered

Class colour

cluster0 cluster1

Status OK

Log x 0

Q5 . Dataset: Mixed Attributes

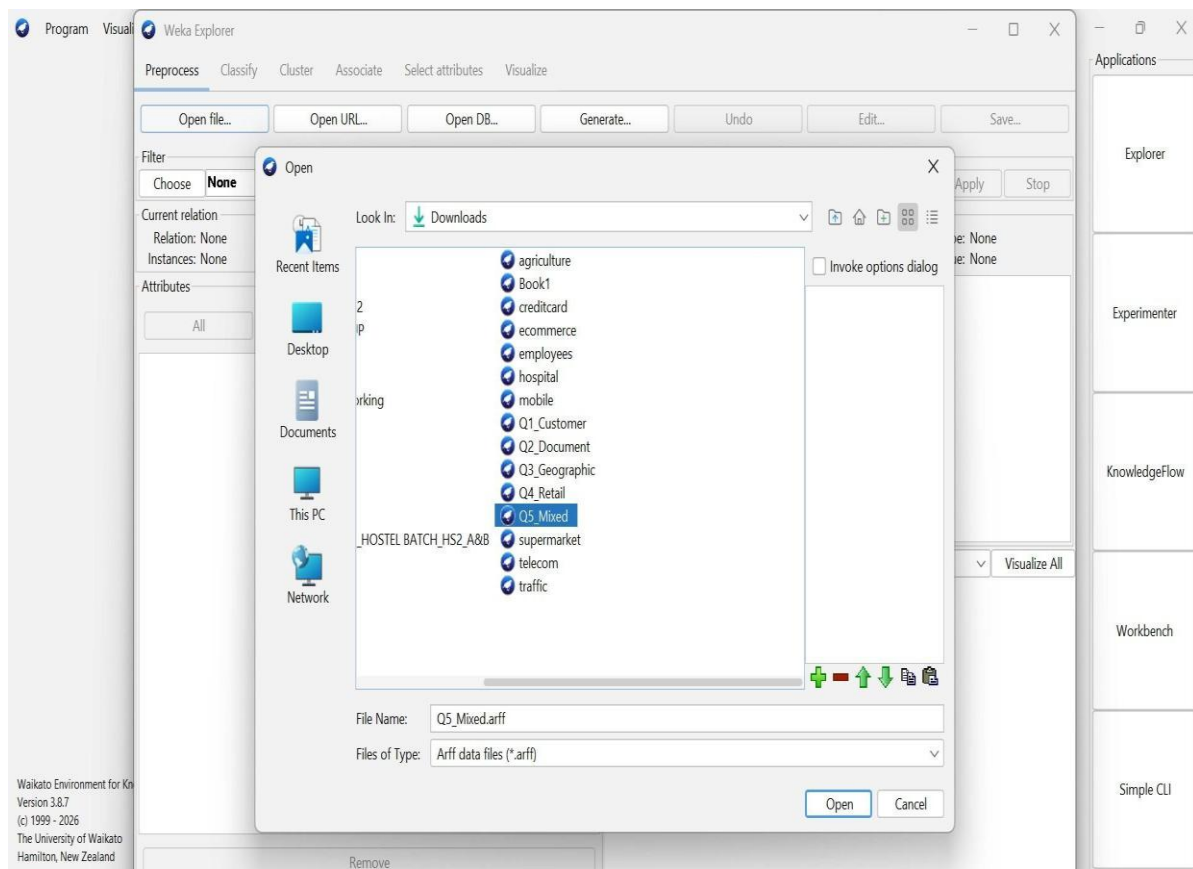
Item Price (₹) Category Rating

I1	500	A	4.5
I2	300	B	3.8
I3	550	A	4.7
I4	250	B	3.5
I5	520	A	4.6

Question:

Compare centroid-based and density-based clustering techniques and analyze their effectiveness on mixed data types. (BL4 – 1 Mark)

OUTPUT:



Weka Explorer

PreprocessClassifyClusterAssociateSelect attributesVisualize

Clusterer

ChooseSimpleKMeans -init 0 -max-candidates 100 -periodic-pruning 10000 -min-density 2.0 -t1 -1.25 -t2 -1.0 -N 2 -A "weka.core.EuclideanDistance" -R first-last" -I 500 -num-slots 1 -S 10

Cluster mode

Use training set

Supplied test set

Percentage split

Classes to clusters evaluation

Store clusters for visualization

Set...

% 66

(Num) Rating

Ignore attributes

Start

Stop

Result list (right-click for options)

06:59:20 - SimpleKMeans

Cluster output

Initial starting points (random):

Cluster 0: 250,B,3.5

Cluster 1: 300,B,3.8

Missing values globally replaced with mean/mode

Final cluster centroids:

Attribute	Cluster#		
	0	1	
Full Data	(5.0)	(2.0)	(3.0)
=====			
Price	424	275	523.3333
Category	A	B	A
Rating	4.22	3.65	4.6

Time taken to build model (full training data) : 0.01 seconds

=== Model and evaluation on training set ===

Clustered Instances

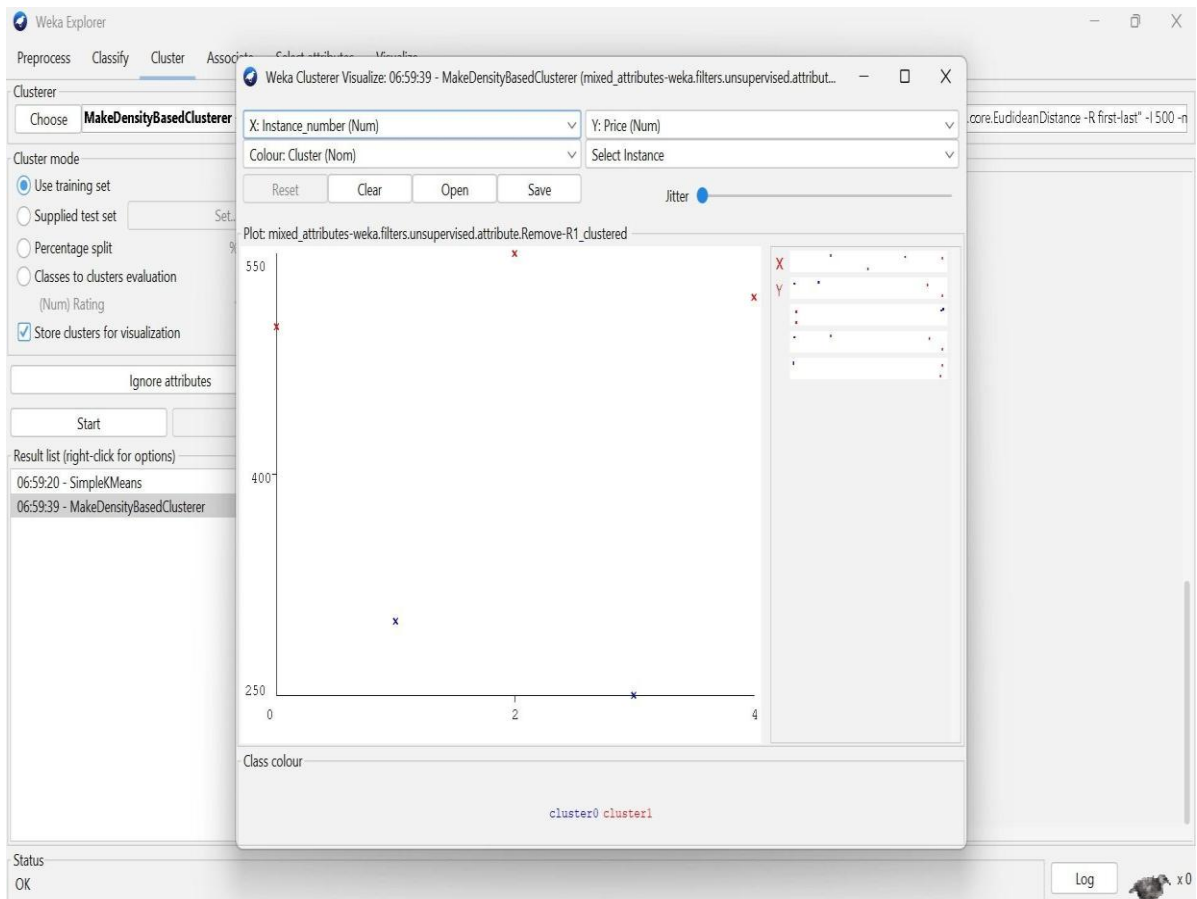
0	2	(40%)
1	3	(60%)

Status

OK

Log

x0



Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Clusterer

Choose **MakeDensityBasedClusterer** -M 1.0E-6 -W weka.clusterers.SimpleKMeans -- -init 0 -max-candidates 100 -periodic-pruning 10000 -min-density 2.0 -t1 -1.25 -t2 -1.0 -N 2 -A "weka.core.EuclideanDistance -R first-last" -I 500 -n

Cluster mode

☒ Use training set

☐ Supplied test set Set...

☐ Percentage split % 66

☐ Classes to clusters evaluation (Num) Rating v

☒ Store clusters for visualization

Ignore attributes

Start Stop

Result list (right-click for options)

06:59:20 - SimpleKMeans

06:59:39 - MakeDensityBasedClusterer

Clusterer output

Normal Distribution. Mean = 2.75 StdDev = 2.5

Attribute: Category

Discrete Estimator. Counts = 1 3 (Total = 4)

Attribute: Rating

Normal Distribution. Mean = 3.65 StdDev = 0.15

Cluster: 1 Prior probability: 0.5714

Attribute: Price

Normal Distribution. Mean = 523.3333 StdDev = 20.548

Attribute: Category

Discrete Estimator. Counts = 4 1 (Total = 5)

Attribute: Rating

Normal Distribution. Mean = 4.6 StdDev = 0.0816

Time taken to build model (full training data) : 0 seconds

=== Model and evaluation on training set ===

Clustered Instances

0	2	(40%)
1	3	(60%)

Log likelihood: -4.6007

Status OK Log x0